

USING R TO AUTOMATE REPORTING

A case study of our work with Salt

What we have achieved

From a presentation that the CHRO of Salt (then Orange) gave in 2014

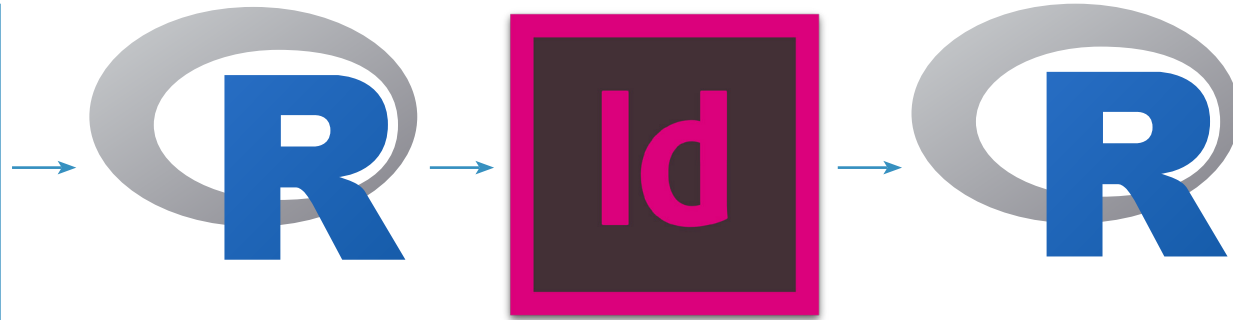
Measurement / KPI	2013	Target	Achieved
Faster report production	1 week	1 day	1 hour
Reduced error rate	10	2 / 3	1
Reduced number of manager enquiries	10	5	2

Our process

R is the core of what we do



Collect data



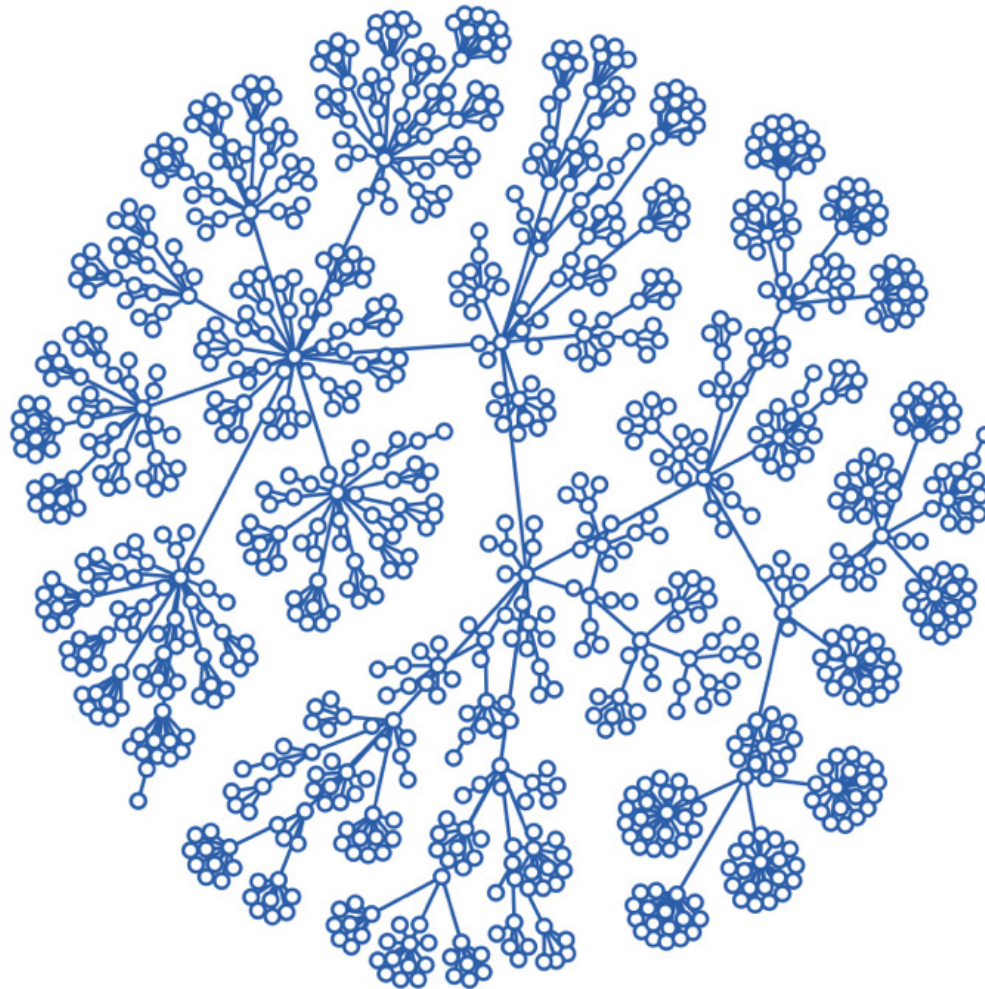
Clean & create
numbers / graphics

layout

file organization

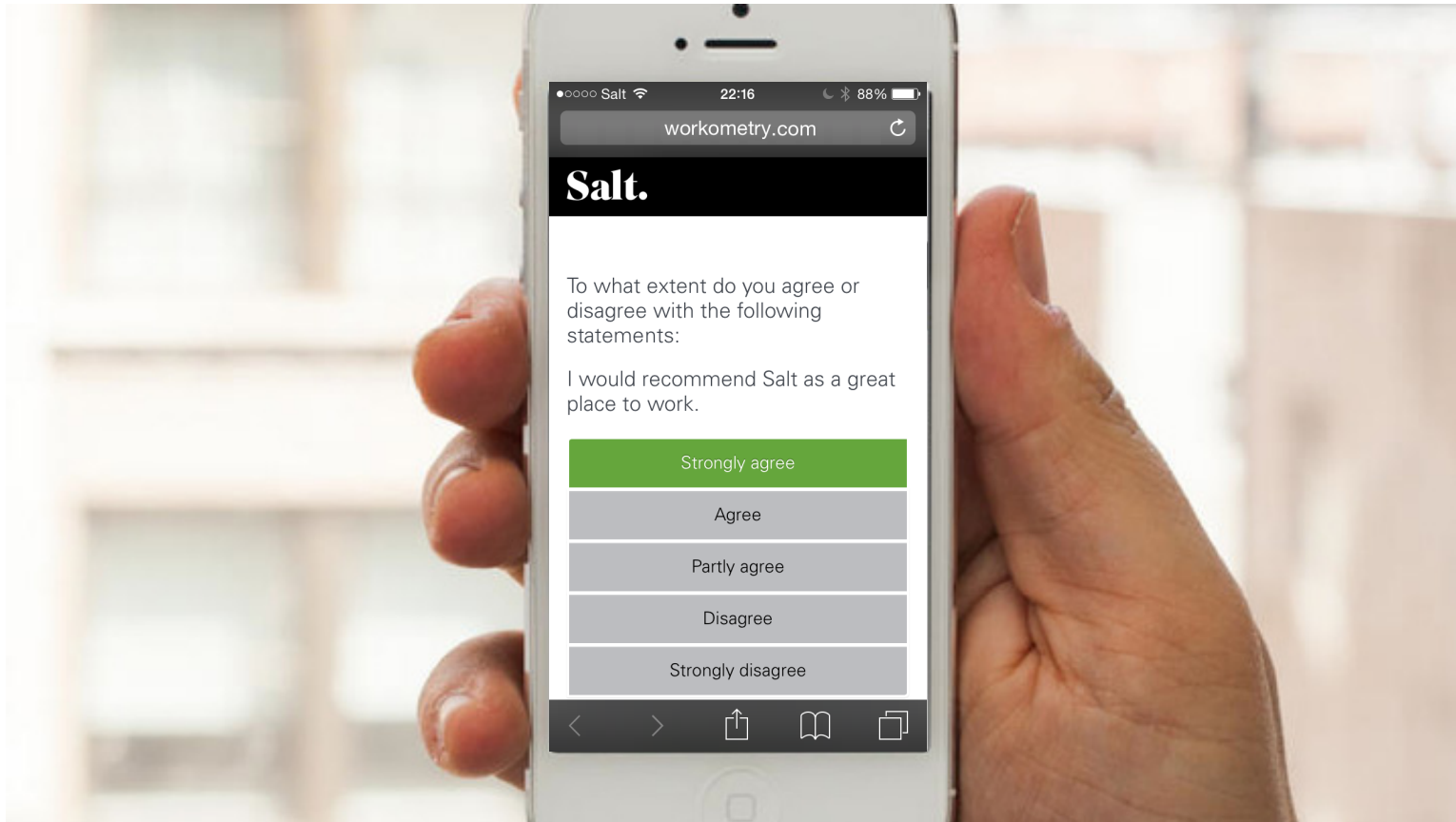
Using igraph to determine relevant teams

We view the organization as a graph (tree). Graph metrics can then be used to identify groups for analysis / reporting purposes. The graph visualisation can be scaled / coloured to show various data such as demographics (age, gender, performance) and survey scores.



Capturing the data

Survey can be used on mobile or desktop devices



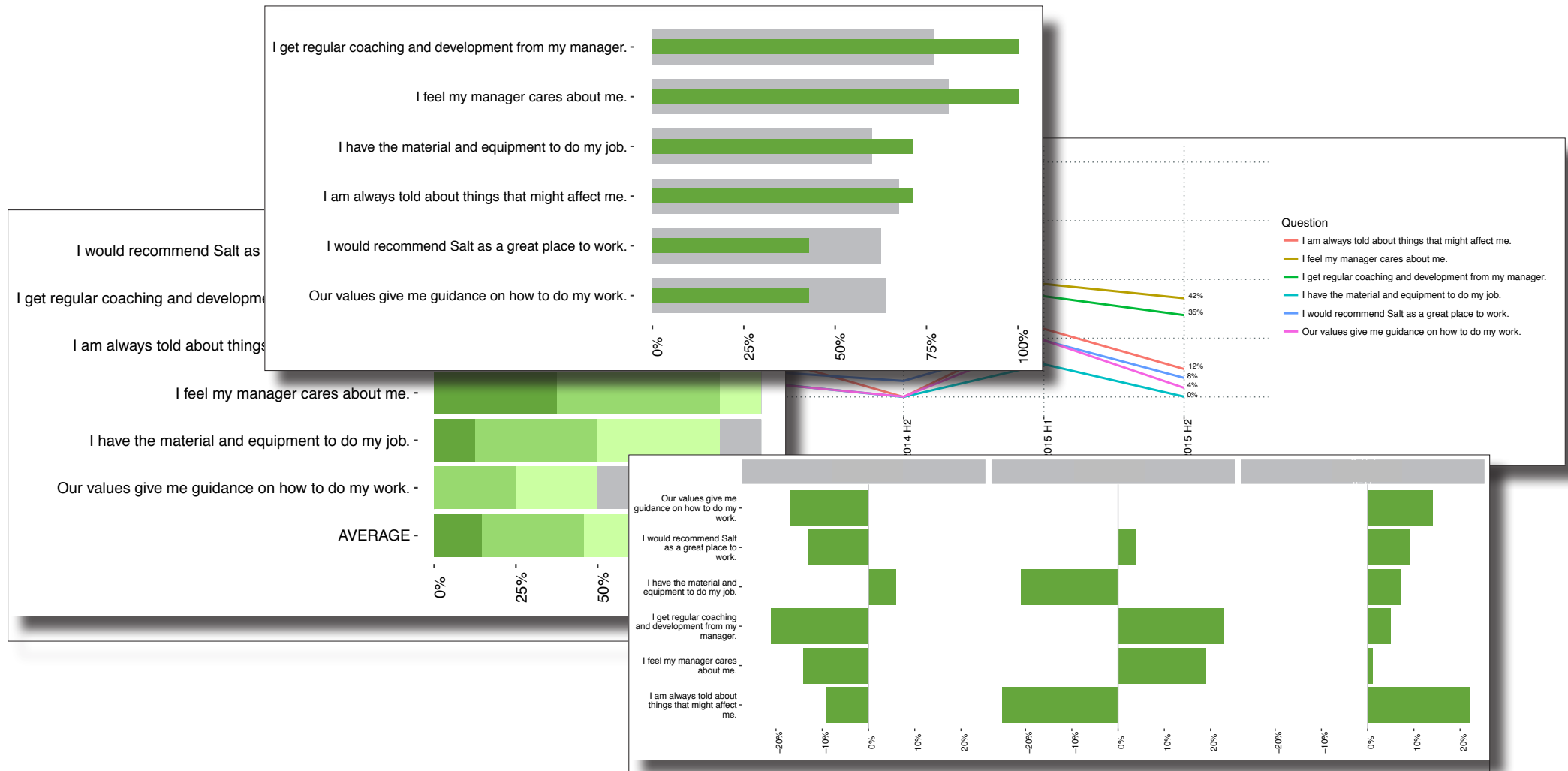
Process within R

High-level approach to the code

```
> Build a tree of the organization (client provides a CSV before survey starts)
> Download survey data from BigQuery
> Do any data cleaning / transformation
> Loop {
>   Identify manager where n=1 neighbourhood >= minimum reporting level
>   Identify from igraph object the employees in his / her team (neighbourhood)
>   filter survey data on these employees
>   Create graphs / numbers needed for report (a function with embedded functions)
>   write line to csv with the numbers / filename
> }
```

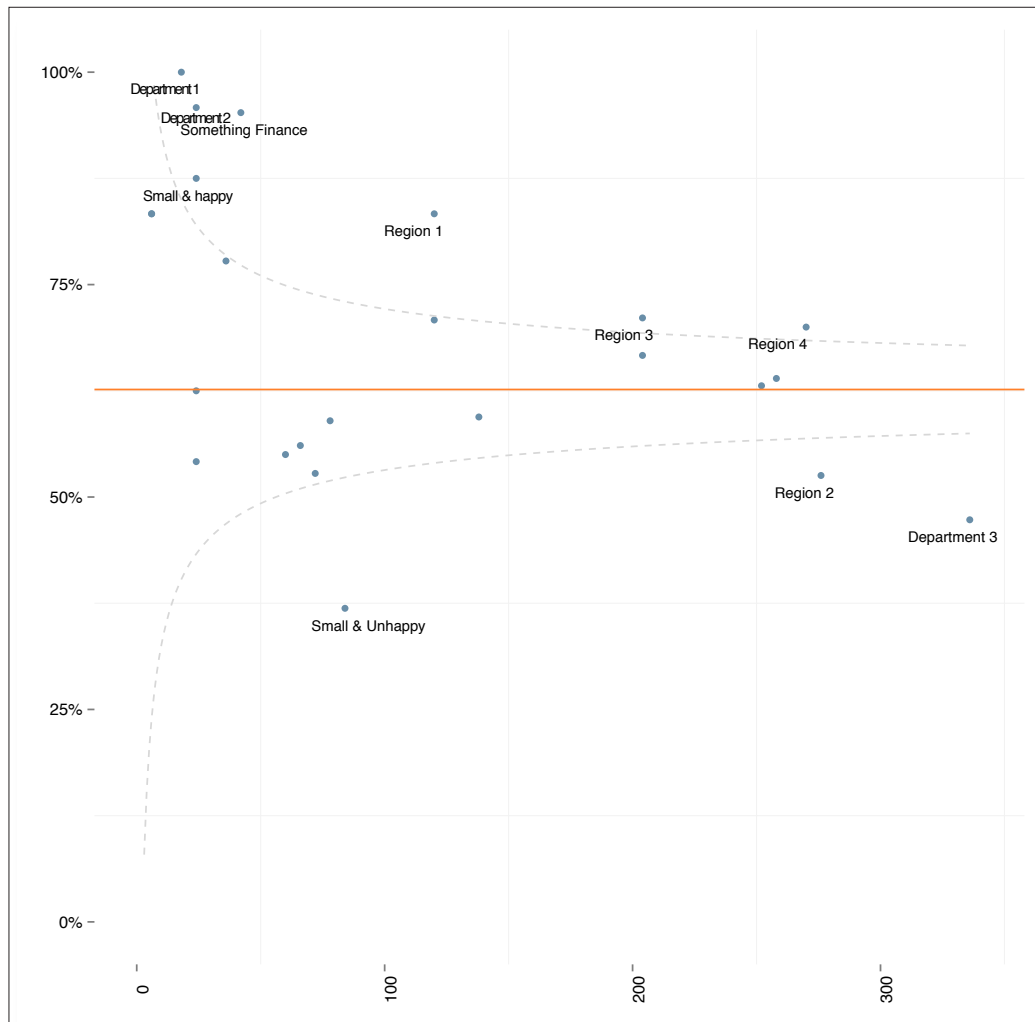
LOTS of graphs produced, but only a few types

We have a reduced set of chart types, each defined within a function. We use dplyr to create the relevant summary table and then send this table to the relevant plotting function. The function returns the ggplot graphic which is then saved.



Clients don't always agree

We wanted to use a funnel plot to show where action was needed - client didn't understand this!

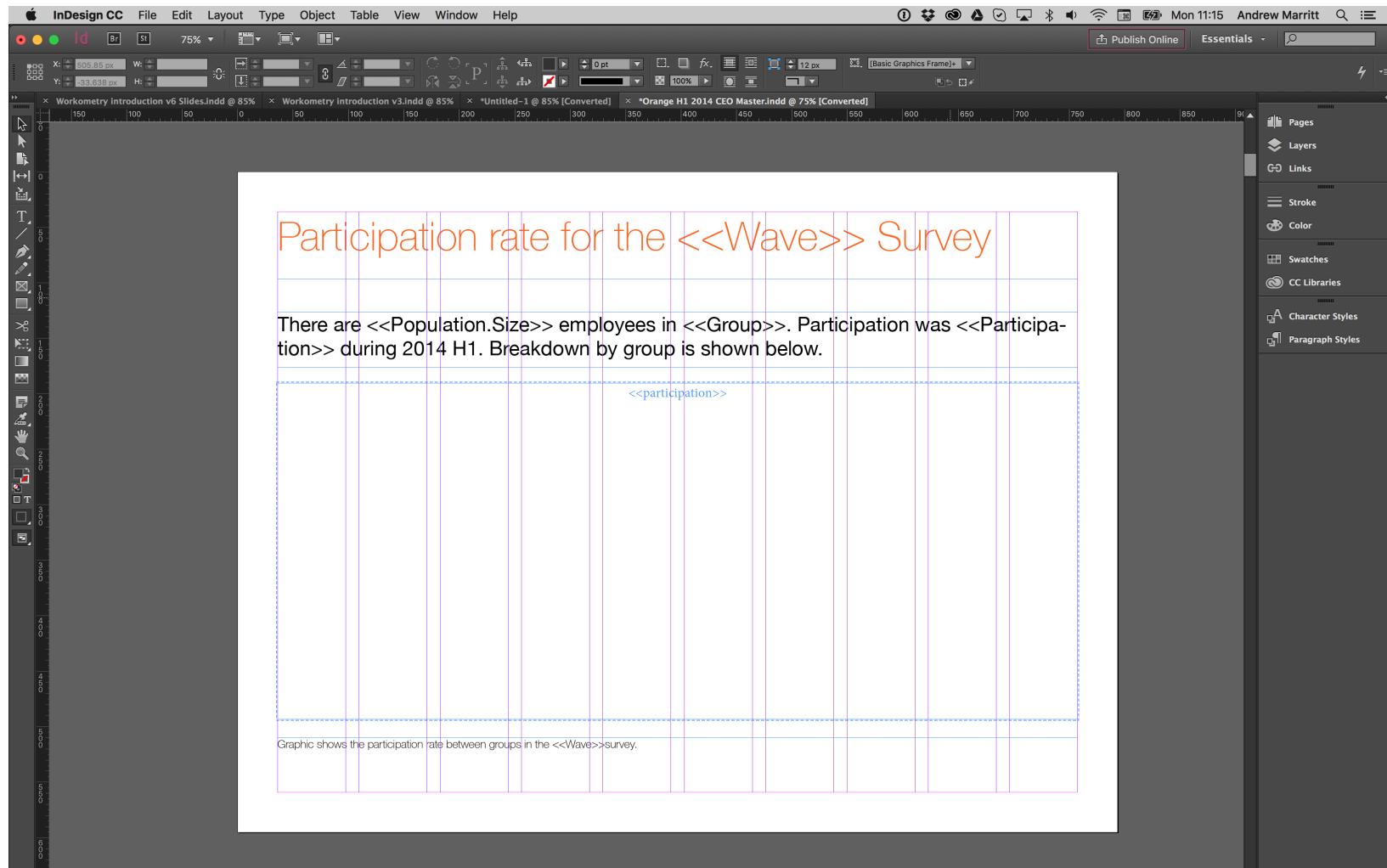


[Funnel plots for comparing institutional performance](#)

David Spiegelhalter, Statistics in Medicine, Statist. Med.
2005; 24:1185–1202

Indesign used to 'mailmerge' data into documents

Input is a csv. Image filename is used, image stored locally.



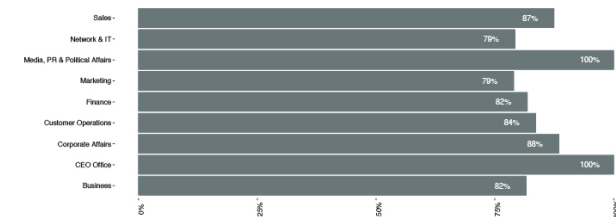
We typically need to produce ~100 reports

Manager reports are based on a standard template. We use Salt's branded design theme (used for external presentations)



Participation rate for the 2015 H2 Survey

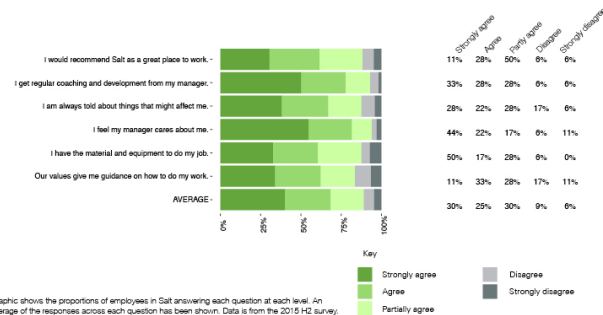
There are 1050 employees in Salt. Participation was 87% during 2015 H2. Breakdown by group is shown below.



Graphic shows the participation rate between groups in the 2015 H2 survey.

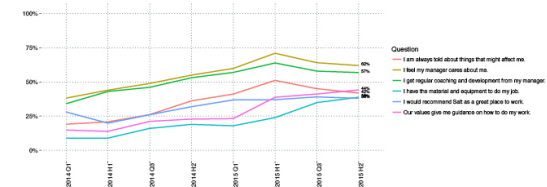
summary of results

The following graphic shows the distribution of responses to all questions during the 2015 H2 survey.



comparison over time - Agree + Strongly

The graphic below shows the difference between Salt responses this time compared to their historic responses.

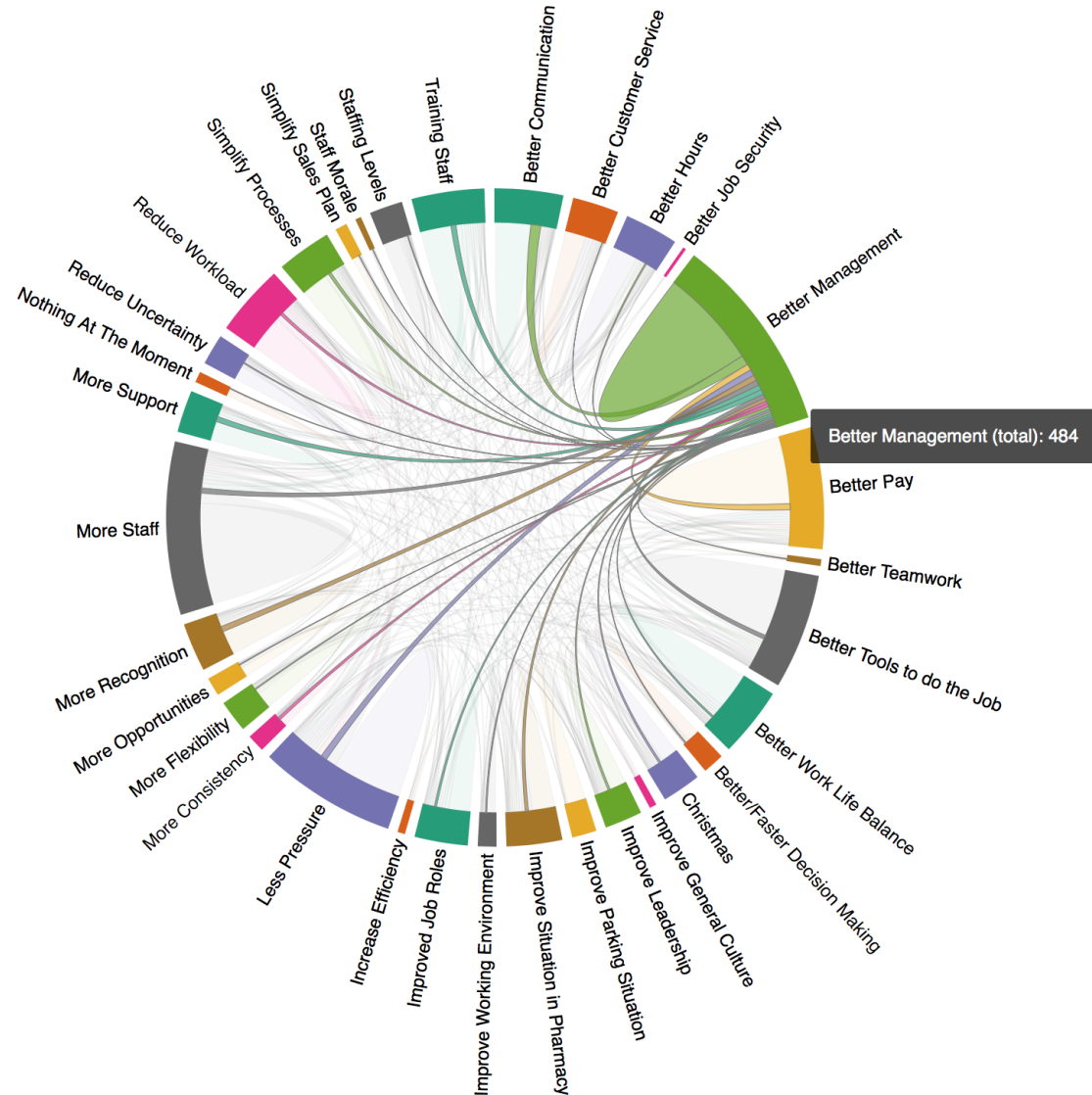


Graphic shows a historic trend of the proportion of responses which were answered as 'Agree' or 'Strongly agree' for Salt.

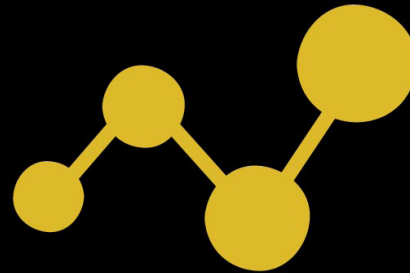
A focus on text - building out Workometry

We believe that understanding text is the key to employee feedback. We prototype in R. Code can be the best way to share specifications with programmers using other .

Chord diagram of topic co-occurrence in a question
created using package [chorddiag](#).



WINNER



PEOPLE ANALYTICS
— AWARDS 2016 —

EMERGING GAME-CHANGER



[See the event report.](#)

Lessons learnt

We've learnt a lot by doing this

- Getting a process production-ready using R is harder than you think
- Data quality is key - always collect data with analysis in mind
- Functions greatly improve the code length but can be harder to debug
- Use styles in ggplot. Consider developing a house style
- R evolves - there is probably a time when you want to re-write code
- Never assume that what you think is intuitive & easy to understand is what your audience will find intuitive & easy

Questions?

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